

Exponential Functions & Their Graphs (3.1)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: \$2,394.43

_____ Given $f(x) = 3^x$, find $f(4)$.

ANSWER: 8

_____ State the horizontal asymptote of $g(x) = -3(2)^x + 7$.

ANSWER: decay

_____ State the horizontal asymptote of $f(x) = 2^x - 5$.

ANSWER: \$1,166.40

_____ For $f(x) = ab^x$, $f(0) = 5$. Find a .

ANSWER: 81 # _____ Is $f(x) = 5(0.7)^x$ exponential growth or decay?	ANSWER: 3 # _____ \$1,000 is invested at 8% compounded annually for 2 years. Find the balance.
ANSWER: (0, 4) # _____ Given $f(x) = \left(\frac{1}{2}\right)^x$, find $f(-3)$.	ANSWER: 5 # _____ (Calculator) \$2,000 is invested at 6% compounded continuously for 3 years. Find the balance to the nearest cent.
ANSWER: $y = -5$ # _____ State the y-intercept of $f(x) = 4(3)^x$.	ANSWER: $y = 7$ # _____ For $f(x) = ab^x$ with $f(1) = 6$ and $f(2) = 18$, find b.